

STATISTICS ^{Assignment} NO.3

1. Compute mean, variance, standard deviation, coefficient of variation, the first four moments about mean, coefficient of skewness and kurtosis for the following frequency distribution. Also comment on the nature of skewness and kurtosis.

x	1	2	3	4	5	6	7	8	9	10
f	7	16	27	35	46	54	41	37	28	9

2. Compute mean, variance, coefficient of variation, coefficient of skewness and kurtosis for the following frequency distribution. Also comment on the nature of skewness and kurtosis.

x	22	24	26	28	30	32	34	36	38	40
f	17	26	37	45	56	74	31	47	38	29

3. Compute mean, median, Q_3 , D_7 , P_{49} , mode, variance, coefficient of variation, coefficient of skewness and kurtosis for the following frequency distribution. Also comment on the nature of skewness and kurtosis.

x	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
f	17	29	34	41	57	63	47	32	17	13

STATISTICS ASSIGNMENT NO.1

1. Define: Variable, discrete and continuous variable with suitable examples.
2. What are the points to be kept in mind while condensing the data in a suitable frequency distribution?
3. Write down the various requisites for an ideal measure of central tendency and dispersion.
4. What are the merits, demerits and uses of Mean, Median and Mode?
5. Compute Mean, Variance, Standard deviation, Coefficient of Variation, the first four moments about mean, Coefficient of Skewness and Kurtosis for the following data.

(a) 7, 9, 7, 5, 7, 3, 4, 9, 2, 7.

(b) 12, 22, 34, 18, 26, 24, 16, 14, 28.

(c) 1.5, 2.3, 3.0, 1.9, 2.6, 2.4, 2.6, 1.6, 1.7, 1.8.

STATISTICS ASSIGNMENT NO.2

1. A distribution consists of three components of sizes 200, 150 and 250 with means 25.5, 31.0, 42.5 with SD's 7.0, 5.0 and 8.0 respectively. Find the combined mean and S.D.
2. For a group containing 100 observations, the arithmetic mean and standard deviation are 8 and $\sqrt{10.5}$. For 50 observations selected from these 100 observations the mean and S.D. are 10 and 2 respectively. Find the arithmetic mean and standard deviation of the other half.
3. The mean and S.D of a characteristic of 100 items were found to be 60 and 10 respectively. At the time of calculations, two items were wrongly taken as 5 and 4.5 instead of 30 and 20. Calculate the corrected mean and corrected standard deviation.
4. The mean and S.D of 20 items are found to be 10 and 2 respectively. At the time of calculations, it was observed that one item 8 was incorrect. Find the mean and S.D, if (i) the wrong item is omitted and (ii) it is replaced by 12.
5. The mean and S.D of 10 values were found to be 9.5 and 2.5 respectively. Later on, an additional observation becomes available. This was 15.0 and was included in the original sample. Find the mean and S.D of the 11 observations.
6. For a set of 10 observations on a variable X following calculations were made:
 $V = (X-30)/2$, $\sum V = -1$, $\sum V^2 = 49$. It was later detected that an observation 40 was misread as 30. Find the correct Coefficient of Variation of X.
7. The mean and standard deviation of weekly wages (in Rs.) are given below for two factories A and B

Factory	No. Of workers	Average weekly wage	S.D. of wages
A	100	Rs. 460.00	50
B	80	Rs.490.00	40

Answer the following:

- (i) Which factory pays large amount as weekly wage?
- (ii) Which factory shows greater variability in the distribution of wages?
- (iii) What is the average and S.D. of all the workers in the two factories taken together?